

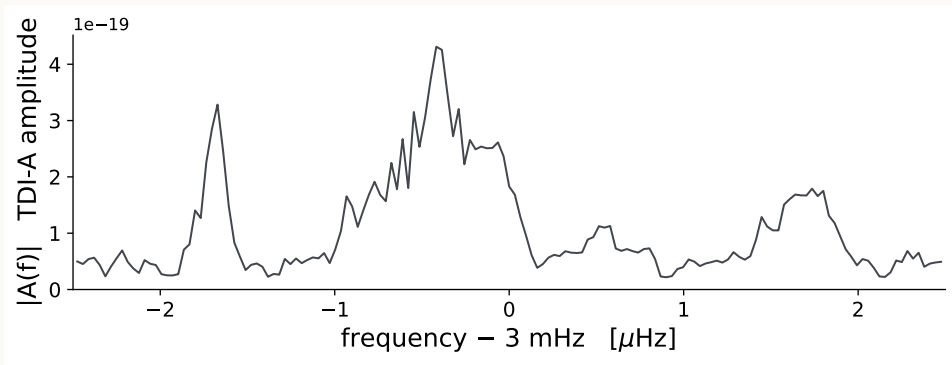
SlotFlow: From Toy Problems to LISA

Two hands-on tutorials on amortized trans-dimensional inference

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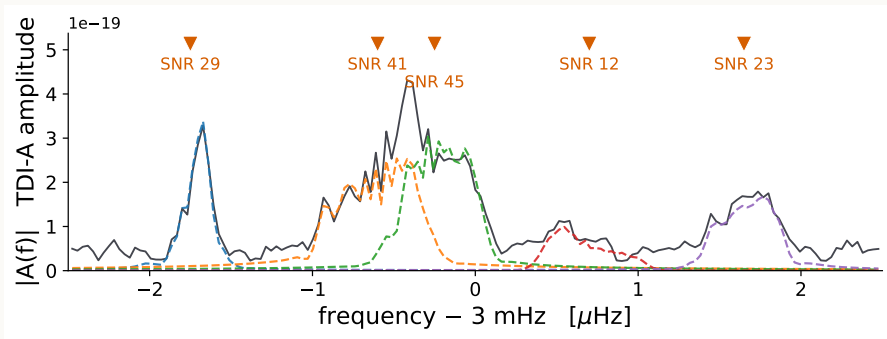
LISA AI Hackathon: Machine Learning Methods for LISA · ESA ESTEC

Warm-up: how many sources are in this signal?



Write down a number — and note how confident you are.

The answer: five — some only visible as overlapping plateaus

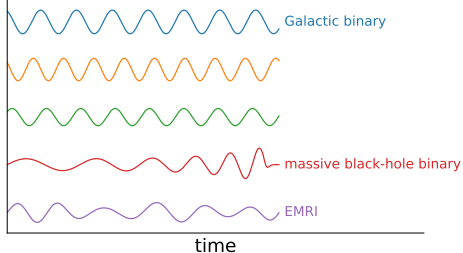


▼ frequency + matched-filter SNR dashed: each injected source alone

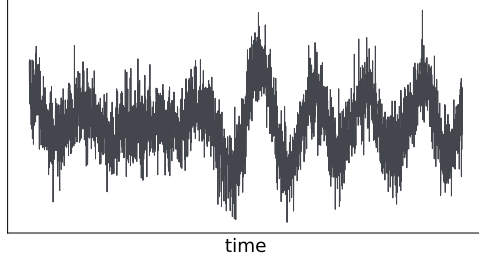
Counting by eye already fails here — a model must infer K *and* the parameters from the mixture alone.

LISA is a catalogue problem: many sources, one stream

many sources, many kinds (schematic) ...



... one noisy observed data stream



$\sim 10^4$ Galactic binaries + MBHBs, EMRIs, backgrounds · no one-event-at-a-time segmentation · calibrated posteriors required

There is no single event to fit — the deliverable is the whole catalogue.

The target is an unordered set — and its size is part of the answer

$$y = \sum_{k=1}^K s(\theta_k) + n, \quad K \text{ unknown}$$

a set whose size is itself unknown — infer $p(K, \{\theta_1, \dots, \theta_K\} \mid y)$

order is meaningless

“source 3” is not a thing;
relabeling changes nothing

overlap blurs resolvability

near the resolution limit:
one source, or two?

K is a parameter

the *dimension* of the
problem is uncertain too

Three structural difficulties — each one returns in the tutorials.

Sampling: costly to rerun — amortization: fixed-dimensional

global trans-dimensional fit

principled reference

~days of sampling,
rerun per dataset & prior

standard amortized posterior

milliseconds per signal

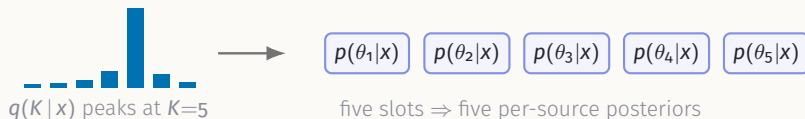
fixed dimension:
 K is a separate step

missing capability

infer K and all θ_k
jointly, in one pass

SlotFlow targets the third column — as a **complement** to the global fit (re-analysis, proposals, triage), not a replacement.

SlotFlow: encode once, count, instantiate, share one flow



Only the encoder knows the physics; counting, slots, and the flow are shared. Cost grows $\sim$ linearly with $\hat{K}$.

Sets have no first element — so training needs assignment

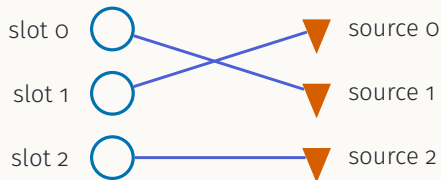
the same two-source answer, written twice

order A: $[\theta_{\text{low}}, \theta_{\text{high}}]$

index-wise loss:
small ✓

order B: $[\theta_{\text{high}}, \theta_{\text{low}}]$

index-wise loss:
huge ✗

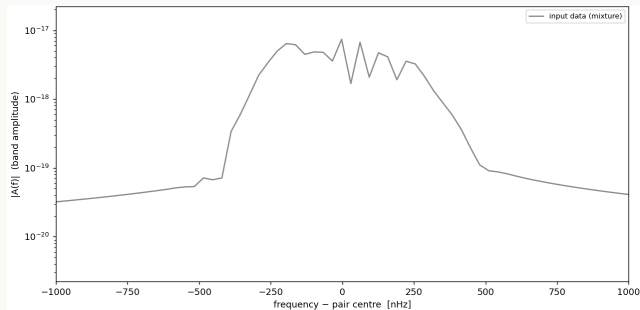


compare every slot to every source;
keep the best one-to-one match

The assignment, not the slot index, defines who is compared with whom — for the **loss** and for **evaluation**.

Tutorial 1: break a model with the index-wise loss — on purpose — then fix it.

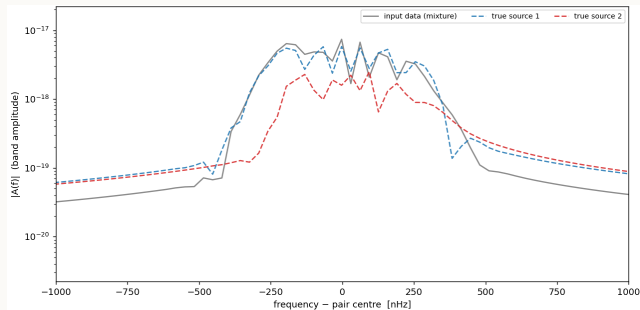
Sometimes even the number of sources is not obvious



one LISA Galactic-binary band spectrum — the mixture is **all the model sees**

What would you report: one broad source — or two overlapping sources?

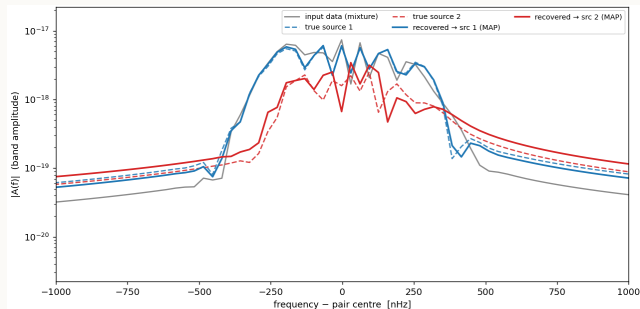
This mixture contains two sources only 17.1 nHz apart



— observed mixture - - / - - the two injected sources

Two sources this close no longer look like two peaks — one source or two becomes a genuine inference problem.

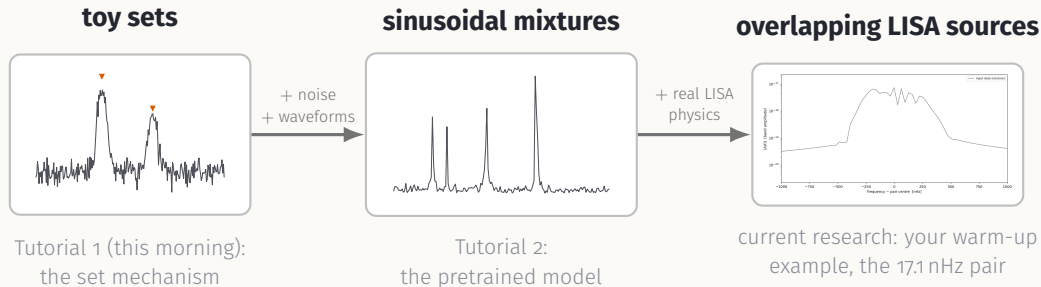
Current SlotFlow recovers both overlapping sources



$K_{\text{true}} = 2$ $\hat{K} = 2$ dashed = truth solid = recovered (MAP)
src 1: $\Delta f = +2.6$ nHz, sky 1.6° **src 2:** $\Delta f = +5.7$ nHz, sky 2.9°

One representative success in the overlapping regime — not a claim that every confused pair is solved.

One problem, three levels of realism



Left to right the encoder and physics become more realistic — the central set-inference problem never changes.

The tutorials teach the basics you need to confront the LISA GB problem.

The two tutorials form one progression

Tutorial 1 — build the mechanism

09:50–10:30 · toy spectral sources

- infer a deliberately simplified toy catalogue;
- deliberately **break** ordered training;
- **one coding TODO**: Hungarian matching.

Tutorial 2 — interrogate the model

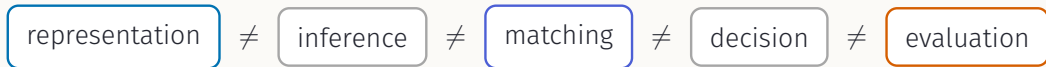
11:00–12:00 · the pretrained SlotFlow

- inspect $q(K|x)$ and candidate slots;
- diagnose **compound failures**;
- **one coding task**: a better catalogue decision rule;
- separate detection from calibration.

One coding task per session · CPU-only, local or Colab · executed solutions released at each session's end.

First you build the mechanism — then nothing about the real model is magic.

One discipline to carry through the morning



Today's failures are engineered *on purpose* — the skill we practice is naming **which layer** failed, instead of letting one layer take the blame for another.

And remember the warm-up: your number, and how unsure you were.

By noon you will know how a model turns that into a calibrated answer — and how to check it.

Set up now

TODO: workshop link / QR

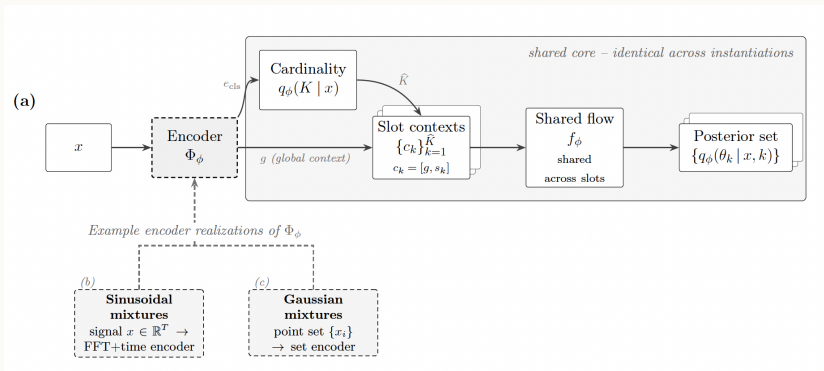
Colab link or the `slotflow_tutorials` folder — both work the same

1. open `slotflow_tutorial_1_toy.ipynb`;
2. run the first cell — it fetches everything it needs;
3. no GPU, no LISA software stack.

Optional live-model checkpoint (464 MB): command in the README — coffee-break wifi, not now.

Backup

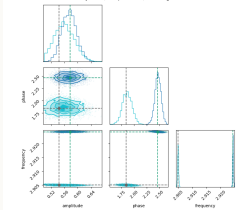
Backup: the architecture, as published



The cardinality head $q_\phi(K | x)$ instantiates $\hat{K}$ slot contexts $c_k = [g, s_k]$; one shared conditional flow yields the posterior set. [arXiv:2511.23228](https://arxiv.org/abs/2511.23228)

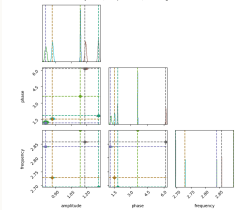
Backup: quantitative results (held-out mixtures)

Posterior over Parameters by Slot (Sample 4029) - data generation in inference mode



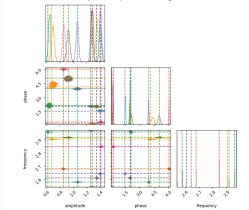
$K = 2$

Posterior over Parameters by Slot (Sample 1898) - data generation in inference mode



$K = 5$

Posterior over Parameters by Slot (Sample 3022) - data generation in inference mode



$K = 10$

- counting: **99.85%** correct over 10^4 held-out signals;
- vs. RJ-MCMC: amplitude/phase posteriors match (Wasserstein $< 0.01/0.03$); frequency **2–3× broader**;
- calibrated (simulation-based calibration); $\sim$ ms per signal $\approx 10^6\times$ faster than the sampler.

Backup: the flow, in one paragraph

- one conditional normalizing flow, **shared across slots**: alternating rational-quadratic-spline and affine autoregressive transforms (nflows), $\sim 26\text{--}31\text{M}$ parameters;
- slot context $\mathbf{c}_k = [\mathbf{g}, \text{onehot}(\mathbf{k})]$ — slot k 's posterior is identical no matter how many other slots exist;
- training: Hungarian-matched flow NLL — assignment on a *detached* cost matrix, loss gathered from the *live* one (Tutorial 1's TODO, with squared error in place of NLL);
- circular phase enters as $(\cos \varphi, \sin \varphi)$.

Backup: how the sandbox maps to LISA (and what is missing)

	tutorial sandbox	real Galactic binaries
source model	sinusoid ($\mathbf{a}, \varphi, \mathbf{f}$)	quasi-monochromatic GW ($\mathcal{A}, \varphi_0, \mathbf{f}, \dot{\mathbf{f}}, \dots$)
band	2.5–3.0 Hz (scaled)	mHz
crowding	$\Delta \mathbf{f} \gtrsim 2 \times$ Rayleigh	confusion foreground
noise	interpolated white-ish	instrument + confusion PSD, TDI
response	none	full LISA response

Every omission is named on purpose: $\dot{\mathbf{f}}$, the response, TDI noise, and inter-source correlations are the roadmap to a real instantiation.

Backup: references

- **SlotFlow** — Houba, Giarda, Speri, arXiv:2511.23228 · github.com/nhouba/slotflow-inference
- **Erebor global fit** — Katz et al., PRD 111, 024060 (2025), arXiv:2405.04690
- **LISA** — Definition Study Report, arXiv:2402.07571
- amortized / simulation-based inference for GWs: DINGO-family, NPE for Galactic binaries, population-level SBI — pointers in the break.